

ARYAN BABURAJAN

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EXPERIENCE

Frappe.io Intern

Jan – Apr 2026

- Wrote a proof-of-concept Unicorn worker using Python 3.14 Subinterpreters, enabling true concurrent parallel execution beyond the limitations imposed by the GIL.
- Integrated Bloomberg's Memray profiler into Frappe's web dashboard to support memory and performance analysis.
- Extended Redis Queue's WorkerPool to allow workers to process multiple queues concurrently, improving memory usage by 32%.

Ducktape Creator

2021 – Present

- Built a custom 3D game engine in C++ with ECS-based architecture using EnTT.
- Implemented a real-time renderer in OpenGL, with prior Vulkan rewrites exploring rendering performance and architecture.
- Added native C++ DLL scripting support to enable modular engine extensions.
- Developed an ImGui-based editor for in-engine debugging and content creation workflows.

SerenityOS Contributor

Mar 2024

- Improved Unicode support for several native applications and learned about character encoding standards.
- Ported applications to use SerenityOS's GML GUI markup language.

Game Development Workshop Mentor Tutor

2025
Thrissur, India

- Mentored 50+ students in an introductory Unity game development workshop.
- Tutored 20+ students in a beginner game development session using p5.js.

TECHNICAL SKILLS

- **Languages:** C++, C, Python, TypeScript, JavaScript, C#
- **Graphics:** OpenGL, Vulkan, EnTT, Dear ImGui, Unity
- **Web:** React, Next.js, Nest.js, Express.js, Node.js
- **Tools:** Git, CMake, Blender, MySQL, MongoDB

PROJECTS

- Mirage - Personalized Product Hunt-style platform with authenticated dashboards and scheduled email newsletters.
- DirectShare - File transfer service using Express.js, EJS, and Netlify for worldwide peer-to-peer file sharing.
- Plaintext - Minimalist text-only browser written in Python.
- Godspeed - Experimental room-based file sharing platform using WebSockets, Express.js, and React.
- Robot - Web crawler written in Python.
- Burrito - Experimental note-taking app inspired by Notion and Excalidraw with a custom canvas system.
- Chaos Car Destruction - Physics-based destruction simulator developed in Unity.

EDUCATION

Christ College of Engineering

Bachelor of Technology in Computer Science and Engineering

2025 – Present
Kerala, India